

PROBABILITY & DISTRIBUTIONS – COMPLETE EXAM NOTES

PART A – 2 MARK QUESTIONS

1. Probability is a numerical measure of the likelihood of an event. $P(A) = \text{favourable outcomes} / \text{total outcomes}$.
2. Axioms: (i) $P(A) \geq 0$ (ii) $P(S)=1$ (iii) For mutually exclusive events, $P(A \cup B) = P(A) + P(B)$.
3. Conditional probability: $P(A|B) = P(A \cap B) / P(B)$, $P(B) > 0$.
4. Bayes' theorem: $P(A|B) = P(B|A)P(A) / P(B)$.
5. A random variable assigns numerical values to outcomes. Example: number of heads.
6. PMF gives probability of a discrete random variable: $\sum P(x) = 1$.
7. PDF describes continuous variables: $\int f(x) dx = 1$.
8. Distribution function is non-decreasing, $0 \leq F(x) \leq 1$.
9. Moment is $E(X^r)$.
10. MGF: $M(t) = E(e^{tX})$.
11. Binomial distribution: $P(X=x) = {}^nC_x p^x q^{n-x}$.
12. Poisson distribution: $P(X=x) = e^{-\lambda} \lambda^x / x!$.
13. Normal distribution mean= μ , variance= σ^2 .
14. Uniform distribution: $f(x) = 1/(b-a)$, $a \leq x \leq b$.

PART B – 10 MARK QUESTIONS

1. Axioms of probability: stated with proofs and examples.
2. Bayes' theorem derived using conditional probability with real-life medical example.
3. Probability of drawing two red balls without replacement = $5/14$.
4. Random variables: discrete and continuous explained with examples.
5. Mean=1.7, Variance=0.81 for given discrete distribution.
6. Mean and variance of Poisson distribution are both λ .
7. Exponential distribution: $P(X < 400) = 1 - e^{-(0.8)}$.
8. MGF of Normal distribution: $\exp(\mu t + \sigma^2 t^2 / 2)$.
9. Uniform vs Exponential comparison table.
10. For $f(x) = 3x^2$, Mean= $3/4$, Variance= $3/80$.

PART C – 16 MARK QUESTIONS

1. Bayes' theorem proved and factory problem solved. Probability product from Machine A = 0.5.
2. Binomial distribution mean= np , variance= npq , MGF= $(q+pe^t)^n$.
3. Discrete distribution: $k=1/25$, Mean=2.8, Variance=1.36, MGF derived.
4. Normal distribution as limiting case of Binomial. Area between $z=0$ and $z=1.96 = 0.475$.
5. Poisson distribution with $\lambda=2.5$: probabilities for exact and at-most cases derived.